



**higher education
& training**

Department
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

NQF LEVEL 2

**REVIEWED INTEGRATED SUMMATIVE
ASSESSMENT TASK**

PROGRAMME ISAT

INSTRUCTIONS TO STUDENTS

PROGRAMME: MECHATRONICS LEVEL 2

SUBJECT CODES: 10040052; 12040212; 06030092

DATE OF IMPLEMENTATION: 01 JANUARY 2020

This document consists of 8 pages.

This is a practical application task that tests your ability to apply the knowledge and skills acquired in the vocational subject. The work required from you is an important and compulsory component of your studies.

1. General information

- 1.1 Study the task carefully.
- 1.2 Note the mark and time allocations per sub-task and for the task as a whole.
- 1.3 Note the assessment date(s).
- 1.4 Study the resources required carefully.
- 1.5 Use the assessment tool as guidance on how your work will be marked.
- 1.6 You are required to provide proof that you submitted work done by yourself.

2. Task description

The ISAT consists of three sub-tasks. For sub-task 1, you must write and simulate a PLC program and assemble a personal computer. For sub-task 2, you must design and wire an electronic circuit and for sub-task 3, you must analyse a mechanical drawing and manufacture the drawn work piece.

All three sub-tasks require worksheets to be completed. The worksheets will be handed to you on the assessment days only. Make sure that you have a pen and pencil to complete the worksheets and that you hand in the completed worksheets for marking.

3. Instructions for the task

3.1 Sub-task 1: Design a PLC program and assemble a computer

(Total: 50 marks; 180 minutes)

Activity 1.1: Identify computer peripherals (Sub-total: 6 marks; 15 minutes)

During a supervised session, you must use a pen to complete Worksheet 1, handed out at the beginning of the session only.

Activity 1.2: Connect the peripherals to the computer (Sub-total: 9 marks; 45 minutes)

Your lecturer will give you instructions to connect the different peripherals identified in activity 1.1 to the computer tower and boot up the computer.

Activity 1.3: Produce a Word document and complete a PLC wiring diagram
(Sub-total: 15 marks; 50 minutes)

Brief: A normally open pushbutton (*start*) is connected to *input 1* of the PLC, a normally closed pushbutton (*stop*) is connected to *input 2* and relay K1 is connected to *output 1* of the PLC. You must switch the power to the PLC *on* and *off* by means of an *on/off* detent switch.

Operation: When the start pushbutton is pressed, *output 1* will energise and latch through an auxiliary contact of *output 1* and stay *on*. An *on-delay* timer will be energised at the same time. The output will stay *on* for *ten* seconds and switch *off*. The output can also be de-energised by pressing the stop pushbutton *input 2* (*stop*).

Create a material list:

- Create a new document in MS Word.
- In the header of this document, put your NAME, STUDENT NUMBER and the DATE.
- In your document, create a Table for the materials list for this brief.
- Use the example in Figure 1 to draw up the table.
- Use the following specifications:
 - the first row of the table must be double spacing;
 - the heading must be centred;
 - *all* header text must be *Arial 12 bold*.
- Fill in the missing information on your materials list.
- The text must be in *Italic*.
- Save your work on a flash drive.
- Print a copy of your document.

Material List			
Inputs			
Item	Description	Marking	Address
1			
2			
3			
Outputs			
4			
5			
PLC Unit			
Special Relays			

FIGURE 1

Complete a wiring diagram:

Your lecturer will hand out Worksheet 2. Complete the PLC wiring diagram and hand in the completed worksheet for marking.

Activity 1.4: Creating a PLC source program and simulate program
 (Sub-total: 20 marks; 80 minutes)

Brief to students: When I1 (START) is activated, Q1 will be ON and latched through a latching contact of Q1. Simultaneously a timer will be activated for 10 seconds. After the 10 seconds have elapsed, Q1 must switch OFF. Q1 can also be switched OFF when I2 (STOP) is activated any time before the 10 seconds elapsed.

Marking
 11
 12

Please turn over

Note: The inputs to the PLC must be provided by the lecturer.

Use any PLC source code or PLC software to design a PLC program. The program must be simulated and debugged if any faults. You must simulate the program. Upload the program to a PLC. Connect the output from the PLC to the circuit built for the Electro Technology ISAT.

3.2 Sub-task 2: Layout and wiring of electrical/electronic circuit

(Total: 50 marks; 150 minutes)

Activity 2.1: Enclosure layout and mounting of component, knowledge questionnaire
(Sub-total: 15 marks; 30 minutes)

The lecturer will give you a circuit diagram and all components needed for the task. You are required to mark off the layout for the push button, toggle switch and female plugs. You must then drill the holes on the enclosure and mount the components. Make sure that all components are mounted securely and that all work done is neatly.

Finally, you will be issued with a knowledge questionnaire which you must complete and hand back to the lecturer.

Activity 2.2: Draw a layout diagram of the electronic components on the vero board. Assembling and soldering of the electronic components on the vero board.
(Sub-total: 25 marks; 90 minutes)

Your lecturer will hand out Worksheet 4. Study the circuit diagram carefully and note the layout of components and where all connections must be made. Draw the layout of how you will mount the components on the vero board template and show where the track breaks on the board will be. The lecturer will check your completed layout and then supply you with all components needed for the task.

You must then:

- Adhere to workshop health and safety rules and regulations.
- Prepare the workstation for a soldering task.
- Select the correct hand tools required to construct the circuit.
- Select the required components to construct the circuit.
- Refer to the layout diagram and place the components on the vero board.
- Mount and solder the components on the vero board.
- Connect the fly leads from the vero board to the push buttons, toggle switch and female plugs mounted on the enclosure.
- Check the circuit board for incorrect polarities, solder shorts, dry joints and broken tracks

Activity 2.3: Knowledge questionnaire (Sub-total: 10 marks; 30 minutes)

Answer all the questions on Worksheet 5 and hand in the answer sheet to your lecturer.

3.3 Sub-task 3: Manufacture a work piece (G-clamp) (Total: 50 marks; 240 minutes)

You must manufacture a G-clamp from a given scaled engineering drawing.

Activity 3.1: Planning for the manufacturing task (Sub-total: 15 marks; 30 minutes)

- Study the not-to-scale drawing of the G-clamp.

- Answer the planning-related questions on Worksheet 6.

Activity 3.2: Manufacturing the work piece (Sub-total: 20 marks; 180 minutes)

- Adhere to workshop health and safety rules and regulations.
- Study the not-to-scale engineering drawing.
- Select the correct hand tools required to manufacture the work piece.
- Manufacture the work piece.

Note: this activity must be done under supervision and does not need to be completed in one continuous session.

Activity 3.3: Knowledge questionnaire (Sub-total: 15 marks; 30 minutes)

Complete Worksheet 7 under supervision and hand in for marking.

4. Time and mark allocation

Sub-task	Activity	Time allocation	Mark allocation	Assessment tools
1	Designing and simulating of a PLC program and assembling of a personal computer	180 minutes	50	Checklist
2	Layout of electrical and electronic components and wiring of circuit	150 minutes	50	Checklist
3	Manufacturing of a work piece (G-clamp)	240 minutes	50	Checklist
TOTAL		570 minutes	150	

5. Assessment tools

Use the checklist on pages 6 as guidance on how your work will be marked.

STUDENT'S CHECKLIST: INTEGRATED SUMMATIVE ASSESSMENT TASK - SUB TASK 1

Sub-task 1: Design a PLC program and assemble a computer

Identify and connect peripherals to computer chassis and boot up

- Safety - the power was off and unplugged while installing peripherals.
- The peripherals were identified on the questionnaire.
- The peripherals were installed to the correct slots.
- The computer was correctly booted up.

Produce material list in Word format and complete the PLC wiring diagram

- MS Word document was correctly opened.
- Created a table with the correct format and column headings.
- Correctly populated of the materials list.
- Drew a complete electrical circuit on the provided template.

Design and simulate a PLC program

- A PLC program was designed according to the design brief.
- Correct inputs and outputs were used according to brief.
- Simulated and debugged the program.

SUB-TOTAL

50

Sub-task 2: Layout and wiring of electrical/electronic circuit

The layout of enclosure and mounting of components

- Measuring and marking off done correctly.
- Enclosure was secured for drilling.
- Enclosure was drilled to specification. (Holes for push buttons, toggle switch, female connectors and the LED.)
- Components were assembled, fitted to specification and mounted securely.
- The overall manufacturing process was completed according to sequence plan.
- The tools were cleaned and returned to the store.
- Work safety and housekeeping were applied throughout the process.

Knowledge questionnaire

- Five tools were listed.

Vero board layout drawing

- Components were clearly labelled.
- All soldering points were indicated on layout.
- All track breaks on vero board were clearly indicated.
- Clear indication given of where field devices (push buttons, battery, toggle switch) must be connected to vero board.

The layout, correct connection and soldering of components

- All solder joints good.
- All connections to components tight and secure.
- Correct operation of circuit.

Safety and housekeeping

- Correct PPE worn.
- Safety use of tools (soldering iron).
- Work place cleaned and tools stored.

	SECTION MARK	TIME (15)
Knowledge questionnaire	[10]	
• All questions on Worksheet 5 were correctly answered.	10	
SUB-TOTAL	[50]	
Sub-task 3: Manufacture a work piece (G-clamp)		
Planning to manufacture (Knowledge questionnaire)	[15]	
• Listed five processes in the work sequence that will be used to complete the work task.	5	
• Tool requirements.	5	
• Safety requirements.	5	
Manufacturing work piece	[20]	
• The materials were correctly cut to size.	1	
• The work piece was de burred.	1	
• The work piece was filed to reference.	1	
• The work piece was measured and scribed.	1	
• The work piece was punched.	1	
• The drill press was prepared for work.	1	
• The work piece was secured for drilling.	1	
• The work piece was drilled to specification.	1	
• The ends of the screw thread were drilled to specification.	1	
• Safety goggles were worn during drilling operations.	1	
• The work piece was cut to remove excess waste.	1	
• The work piece was filed square where the excess waste was removed.	1	
• The hole for the screw thread was tapped to correct size.	1	
• The work piece was chamfered correctly to specification.	1	
• The screw thread handle was force fitted at one end and a washer was force fitted at the other end.	1	
• The work piece was assembled to specification.	1	
• The overall manufacturing process was completed according to the sequence plan.	1	
• The tools were cleaned and returned to the store.	1	
• Work safety and housekeeping were applied throughout the process.	1	
Knowledge questionnaire	[15]	
• Question 1.	5	
• Question 2.	5	
• Question 3.	5	
SUB-TOTAL	50	
TOTAL	150	

6. RECORD OF PERFORMANCE IN INTEGRATED SUMMATIVE ASSESSMENT TASK			
College			
Campus			
Student's Surname and First Name/s			
Student's ID Number			
Lecturer's Surname and Initials			
Date of conclusion of assessment			
ASSESSMENT GRID			
SUB-TASKS	MARK ALLOCATION	STUDENT'S MARK	MODERATED MARK
Sub-task 1: Design a PLC program and assemble a computer	50		
Sub-task 2: Layout and wiring of electrical/electronic circuit	50		
Sub-task 3: Manufacture a work piece (G-clamp)	50		
Total	150		
Converted to %	100%		

COMPETENCE LEVEL INDICATORS				
Outstanding	Highly competent	Competent	Not yet competent	Not achieved
80-100%	70-79%	50-69%	40-49%	0-39%
5	4	3	2	1
Student's Competence Level				
Student's Signature				
Lecturer's Signature				
Date				
Moderator's Signature				
Feedback to student				